PERSONAL REFLECTION

Who Deserves the Credit When a Machine Makes the Discovery?

A Personal Reflection on the Article *Working with Academic Communities to Build Scientific Understanding*

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Introduction

After reading the article, I found myself focusing less on the scientific discoveries themselves and more on their significance and the questions they raise regarding the future of scientific research. The article presents three results in the fields of mathematics, computer science, and physics, produced by artificial intelligence systems, and OpenAI's attempt to involve the academic community in the processes of verifying, examining, and understanding them.

One of the ideas that particularly interested me was the way OpenAI treats the models' outputs as a form of **"raw research data,"** which requires further work by the scientific community in order to be transformed into knowledge and understanding. In my view, this approach reflects a significant change in the way we are accustomed to thinking about the scientific research process. However, the issue that prompted the most thought for me was in fact **the question of credit** for these discoveries.

Giving Up Academic Credit

One of the principles OpenAI presents in the article is that the company does not seek academic credit for the discoveries produced by its models, and that the researchers participating in the process can act independently and receive recognition for their contributions. On the one hand, I appreciate this approach and the attempt to promote collaboration with the scientific community. On the other hand, I began to wonder **whether giving up academic credit necessarily means giving up credit itself**.

In my opinion, commercial companies have a clear interest in benefiting from recognition for significant achievements associated with them. Ultimately, when an OpenAI model arrives at a solution to a scientific problem that has occupied researchers for decades, it is reasonable to assume that the public will primarily remember the company's name. This does not mean that OpenAI necessarily seeks to claim the discoveries as its own, but I believe it is important to distinguish between **giving up formal academic recognition** and **the public and reputational recognition** that accompanies achievements of this kind.

Furthermore, the article itself acknowledges that selecting the first researchers who will gain access to the results creates information asymmetries, a fact that raises additional questions in my mind regarding the company's influence over the process of scientific recognition.

Who Actually Deserves the Credit?

As I thought more deeply about the subject, I realized that the central question for me is not only who receives the credit, but **who truly deserves it**.

If neither the company nor the human researchers conceived the idea or the solution, why should either of them receive credit for the discovery itself?

In such a case, the system is the one that arrived at the result. OpenAI even notes in the article that the mere fact that a researcher was invited to lead a working group does not indicate that the researcher or their group made the discovery. Although the company is responsible for developing the model, and the researchers contribute to examining, verifying, and understanding the results, these are different contributions, each of which has value in its own right. Therefore, in my opinion, we should distinguish between **those who enabled the discovery**, **those who made it**, and **those who verified its correctness**.

However, it is important to qualify this by noting that the article does not detail the full extent of human involvement in the process. Therefore, my argument mainly concerns cases in which the idea or solution was indeed produced by the model and was not directly dictated to it by a human.

The Counterargument

Alongside this position, I am aware that there is also a significant counterargument. Scientific credit does not merely serve to identify who arrived at a discovery, but also functions as **a social mechanism for granting recognition and assigning responsibility**. Unlike a human researcher, an artificial intelligence system cannot benefit from recognition or bear responsibility in the same way.

However, does this mean that credit should automatically be transferred to the company that developed it?

I do not believe so. In my view, the discovery itself can be attributed to the system without necessarily granting it rights or ownership over the result. **The question of who made the discovery and the question of who is entitled to benefit from its outcomes are not necessarily identical.**

Parents, Teachers, and Lecturers

This thought led me to draw a comparison with the human learning process. A researcher who makes a significant scientific discovery is not born with the knowledge and abilities that enable them to achieve it. Throughout their life, they acquire knowledge from their parents, teachers, lecturers, and researchers who came before them, each of whom contributes to some extent to their development.

However, when that researcher makes a groundbreaking discovery, are those individuals entitled to credit for it?

Generally, the answer is no. We recognize their contribution to the researcher’s education and development, but we do not attribute to them the discoveries that the researcher made independently.

If so, why should the situation be different when it comes to artificial intelligence?

Of course, there are fundamental differences between educating a person and developing an AI system. A company developing a model determines its architecture, selects its training methods, and defines its objectives, whereas a parent or teacher does not exercise the same degree of control over a person’s development. Nevertheless, in my view, the underlying principle remains similar: **creating the conditions and capabilities that make a discovery possible is not necessarily equivalent to making the discovery itself**.

Universities and Research Institutions

Following this line of thought, I also found myself considering the role of universities and research institutions. In many cases, the university is the one that provides researchers with laboratories, equipment, funding, and infrastructure

necessary to conduct research. When a significant discovery occurs, scientific credit is generally given to the researchers themselves, but the university may also benefit from prestige, additional research funding, and sometimes even commercial rights arising from the result. In a sense, this situation resembles the activities of AI companies, which provide the tools and resources that make discoveries possible.

However, does the mere provision of resources also grant ownership over the discovery?

This comparison made me realize that the question is more complex than I initially thought, since even within the traditional research system, there is a relationship between those who fund and enable research and those who may benefit from its results. It is possible, therefore, that **artificial intelligence is not creating an entirely new problem, but rather highlighting a tension that already existed within the research system**.

From Discovery to Understanding

Another thought that arose while reading the article concerns the possible change in the role of human researchers. Over the years, a significant part of scientific research has focused on the attempt to make new discoveries. However, if artificial intelligence systems are able to perform an increasing share of the work involved in discovery in the future, it is possible that fewer researchers will engage in discovery itself, while more researchers will focus on examining, verifying, interpreting, and understanding the results.

In this context, I returned to the idea presented at the beginning of the article, according to which the models' outputs constitute a form of **raw research data** that still requires human involvement in order to become scientific understanding. This approach may even suggest how the role of researchers could evolve in the future. I do not believe that human discoveries will disappear entirely, but they may become less common, and the importance we assign to the different roles in the research process may change accordingly. Perhaps in the future, **the central question will no longer be only who managed to discover something new, but also who managed to understand its meaning, verify its correctness, and use it to advance science**.

Conclusion

Ultimately, the article led me to consider whether we need to reexamine the meaning of scientific credit in the age of artificial intelligence. From my perspective, it is important to distinguish between **the discovery itself**, **the verification of the result**, **the development of the system** that enabled it, and **ownership** of the discovery. There is no requirement that all of these be attributed to the same party. In fact, questions of this kind are already being addressed by legal systems, which have had to consider the possibility of recognizing an AI system as an inventor for the purpose of patent registration.

I am still not certain how credit and rights should be distributed in such cases, but I believe that merely developing a system capable of making discoveries does not automatically grant ownership over every result it produces.

The Open Question

Perhaps the most significant question that remained with me after reading the article is one that may become very real in the future:

> **If one day an artificial intelligence system independently arrives at a discovery that enables the development of a drug that saves millions of human lives, would the company that developed and trained the system be entitled to register a patent and profit from the discovery? Does the mere fact that it created the conditions that made the discovery possible also grant it ownership over the result, even though the company itself did not conceive the idea?**

This reflection expresses my personal opinions, thoughts, and insights developed through reading the article, alongside my personal experience as a Computer Science student at the beginning of my professional and research journey. The arguments and predictions presented here reflect my own interpretation and do not necessarily represent the conclusions of the article's authors.